MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 04** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 9: Some Applications of Trigonometry** | **MM:** 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason

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Q1. A car is moving away from the base of a 30 m high tower. The angle of elevation of the top of the tower from the car at an instant when the car is $10\sqrt{3}$ m away from the base is: [1]

(A) 30° (B) 45° (C) 60° (D) 90°

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C)

Justification: $\tan\theta = \dfrac{30}{10\sqrt{3}} = \sqrt{3} = \tan 60^\circ \implies \theta = \boxed{60^\circ}$.

Q2. A kite is flying at a height of 60 m above the ground. The string attached to the kite is temporarily tied to a point on the ground inclined at 60°. Assuming no slack, the length of the string is: [1]

(A) $40\sqrt{3}$ m (B) $30\sqrt{3}$ m (C) $20\sqrt{3}$ m (D) 120 m

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: $\sin 60^\circ = \dfrac{60}{L} \implies L = \dfrac{120}{\sqrt{3}} = \boxed{40\sqrt{3}\text{ m}}$.

Q3. If the height of a vertical pole is $\sqrt{3}$ times the length of its shadow on the ground, then the angle of elevation of the sun is: [1]

(A) 30° (B) 45° (C) 60° (D) 75°

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C)

Justification: $\tan\theta = \dfrac{\sqrt{3}s}{s} = \sqrt{3} \implies \theta = \boxed{60^\circ}$.

Q4. An observer 1.5 m tall is 28.5 m away from a tower 30 m high. The angle of elevation of the top of the tower from the eye of the observer is: [1]

(A) 30° (B) 45° (C) 60° (D) 90°

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B)

Justification: Height above eye level $= 30 - 1.5 = 28.5$ m. Hence $\tan\theta = \dfrac{28.5}{28.5} = 1 \implies \theta = \boxed{45^\circ}$.

Q5. **Assertion (A):** The angle of depression of an object viewed into a valley from a cliff is numerically equal to the angle of elevation of the cliff viewed from that object.
Reason (R): The line of sight acts as a transversal line intersecting two parallel horizontal planes, making alternate interior angles equal. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: Because horizontal eye-level lines are parallel, the transversal line of sight creates equal alternate interior angles. Both (A) and (R) are true and (R) explains (A).

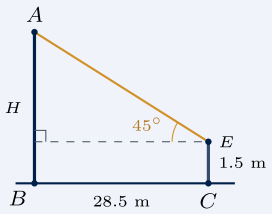
SECTION B: Short Answer Questions

MATHS MANTHAN | (11 Marks)

Q6. An observer 1.5 m tall is 28.5 m away from a chimney. The angle of elevation of the top of the chimney from her eyes is 45°. What is the height of the chimney? [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Let chimney be AB, observer be $EC = 1.5$ m at horizontal distance $BC = 28.5$ m, with eye-level angle 45° above horizontal. [0.5 Mark]



Height above eye level $= h' = 28.5\tan 45^\circ = 28.5 \times 1 = 28.5$ m. [1 Mark]

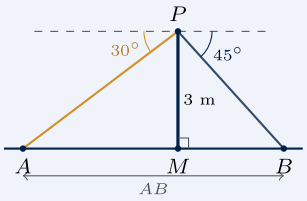
Total chimney height $AB = 28.5 + 1.5 = \boxed{30 \text{ m}}$

Adding eye height: [0.5 Mark]

Q7. From a point on a bridge across a river, the angles of depression of the banks on opposite sides of the river are 30° and 45°, respectively. If the bridge is at a height of 3 m from the banks, find the width of the river. **[3]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let bridge point P be at vertical height $PM = 3$ m above river surface AB, with opposite bank depressions $\angle A = 30°$ and $\angle B = 45°$. [0.5 Mark]



In right $\triangle PMA$: $\tan 30° = \dfrac{3}{AM} \implies AM = 3\sqrt{3}$ m. [1 Mark]

In right $\triangle PMB$: $\tan 45° = \dfrac{3}{MB} \implies MB = 3$ m. [1 Mark]

Width of the river $AB = AM + MB = 3\sqrt{3} + 3 = 3(\sqrt{3}+1)$ m.

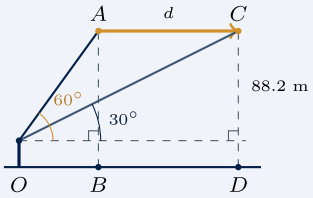
$$AB = 3(1.732 + 1) = 3 \times 2.732 = \boxed{3(\sqrt{3}+1) \text{ m} \approx 8.196 \text{ m}}$$

Final river width: [0.5 Mark]

Q8. A 1.2 m tall girl spots a balloon moving with the wind in a horizontal line at a constant height of 88.2 m from the ground. The angle of elevation of the balloon from the eyes of the girl at any instant is 60°. After some time, the angle of elevation reduces to 30°. Find the distance travelled by the balloon during the interval. **[3]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let balloon move horizontally from A to C at ground height 88.2 m, so vertical height above the 1.2 m girl's eyes (O) is $h = 88.2 - 1.2 = 87$ m. [0.5 Mark]



Let horizontal distances from eye level be x_1 (at 60°) and x_2 (at 30°):

$$x_1 = \frac{87}{\tan 60°} = \frac{87}{\sqrt{3}} = 29\sqrt{3} \text{ m}, \qquad x_2 = \frac{87}{\tan 30°} = 87\sqrt{3} \text{ m}$$

Both horizontal distances: [1.5 Marks]

Distance travelled by balloon $= AC = x_2 - x_1 = 87\sqrt{3} - 29\sqrt{3} = 58\sqrt{3}$ m.

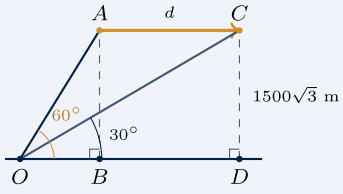
$$d = 58 \times 1.732 = \boxed{58\sqrt{3} \text{ m} \approx 100.456 \text{ m}}$$

Final distance: [1 Mark]

Q9. An aeroplane flying horizontally at a constant height of $1500\sqrt{3}$ m above the ground is observed at an elevation of 60°. After 15 seconds, the angle of elevation is observed to be 30°. Find the speed of the aeroplane in km/h. **[3]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let aeroplane move horizontally from A to C at height $1500\sqrt{3}$ m, observed from O with elevations 60° and 30°. [0.5 Mark]



In $\triangle ABO$: $OB = \dfrac{1500\sqrt{3}}{\tan 60°} = 1500$ m. In $\triangle CDO$: $OD = \dfrac{1500\sqrt{3}}{\tan 30°} = 4500$ m. [1 Mark]

Distance flown in 15 s: $BD = 4500 - 1500 = 3000$ m. [1 Mark]

$$\text{Speed} = \frac{3000}{15} = 200 \text{ m/s} = 200 \times \frac{18}{5} = \boxed{720 \text{ km/h}}$$

Final speed in km/h: [0.5 Mark]

Q10. CASE STUDY: The Iconic Pitampura TV Tower (Doordarshan Delhi)

A surveying student standing at ground point A observes the top (T) of the Pitampura TV Tower at an angle of elevation of $60°$. A technical gallery C lies 78 m directly above tower base B, observed from A at an angle of elevation of $30°$.

(i) Find the horizontal distance between the observation point A and the base of the TV tower (B). [1]

(ii) Find the direct line-of-sight distance from point A to the observation gallery at point C. [1]

(iii) Calculate the total height of the Pitampura TV Tower (BT). [2]

OR

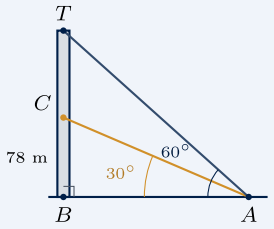
Find the vertical distance between the observation gallery C and the top of the tower T. [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]

(i) In right $\triangle ABC$ ($\angle CAB = 30°, BC = 78$ m):

$$\tan 30° = \frac{BC}{AB} \implies AB = 78\sqrt{3}$$

$$\implies AB = 78(1.732) = \boxed{78\sqrt{3} \text{ m} \approx 135.10 \text{ m}}$$



Subpart (i): [1 Mark]

(ii) In right $\triangle ABC$:

$$\sin 30° = \frac{BC}{AC} \implies \frac{1}{2} = \frac{78}{AC} \implies AC = \boxed{156 \text{ m}}$$

Subpart (ii): [1 Mark]

(iii) Main Choice: In right $\triangle ABT$ ($\angle TAB = 60°$), $\tan 60° = \frac{BT}{AB}$. [1 Mark]

$$\implies BT = AB\sqrt{3} = (78\sqrt{3})\sqrt{3} = 78 \times 3 = \boxed{234 \text{ m}}$$

Total height BT: [1 Mark]

(iii) OR Choice: Since total height $BT = 234$ m and gallery height $BC = 78$ m, [1 Mark]

$$\text{Vertical distance } CT = BT - BC = 234 - 78 = \boxed{156 \text{ m}}$$

Vertical distance CT: [1 Mark]

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★